CMS stress diagnostic: closed outcomes and limits

Private prototype; no production promotion. The source-derived Fortran invariant certificate has conditional QP pole support at the forward CMS fixture. Its complete stress comparison remains failed and incomplete; the finite term remains unresolved. These observations do not qualify normal MadLoop stability/rescue or precise performance.

The fixture is $e^+e^- \to \mu^+\mu^-$ with QED virtual corrections and the existing complex-mass Gmu model. The prototype changes isolated generated Fortran; the original TIRust shared library is unchanged. Fixed mode 4 returns 120, an unchecked return. The eligible workers have both post-call software failure flags clear; this supplies no independent host accuracy guarantee.

Table 1: Execution completeness, eligibility and numerical agreement are separate outcomes.

Closed stage	Outcome	Scope retained
First build attempt	Failed	Controller `KeyError('size')`; zero native commands, loader queries or amplitudes.
Successor ordinary	8 / 8 eligible	Point 1, original/candidate × CutTools/TIRust × DP/QP. Ordinary target met; private fixed-mode evidence only.
Full stress	Incomplete; exit 1	Eight point-20 workers complete. Ninth worker, original CutTools DP at point 21, times out after 300 s, is terminated by signal 15 and reaped. Seven workers never launch.
Point-20 subset	5 / 8 eligible	Three failed TIRust workers' buffers excluded. Candidate TIRust QP is eligible but its comparison with CutTools QP fails; no complete stress result exists.
Legacy reference probe	0 / 2 eligible	NinjaRS and Ninja complete their protocols; both fail their scientific eligibility gates. No coefficient comparison, including bootstrap, is interpreted.

Original/candidate CutTools controls agree exactly. Original TIRust DP/QP and candidate TIRust DP retain software failures and W03 warnings at point 20. Candidate TIRust QP has clear post-call flags but differs from CutTools, with maximum recorded `abs(a-b)/max(abs(a),abs(b))` of 1.237729 across the compared coefficients. The later universal-pole audits below support its poles under their reconstruction assumptions; they do not turn that failed full-coefficient comparison or the incomplete stage into a pass.

Table 2: Both probes use the legacy kinematic helper; matched native input geometry with the prototype is unestablished.

Legacy QP probe	Eligible	Why no coefficient arithmetic is allowed
NinjaRS (8)	No	Five point-20 box groups return status 1: 3, 4, 8, 12, 24. The software QP failure flag is set.
Ninja (6)	No	Adapter statuses are zero but final finite and both pole words are nonfinite. The host software DP failure flag is set despite zero observed DP adapter calls.

Their 104 completed adapter calls establish the QP interface route (storage width 128, precision digits 113), not every reducer's internal working precision. No scalar cutoff was changed and no particular scalar-call cause is proven.

Independent universal-pole comparisons

For four massless unit-charge external fermions, the double-pole prediction is $A_{-2} = -4B\frac{\alpha}{2\pi}$. The source-derived single-IR-pole prediction is $A_{-1} = B\frac{\alpha}{2\pi}\left[-6 - 2\sum_{i<j} Q_i Q_j \ln\left(\frac{|2k_i\cdot k_j|}{\mu^2}\right)\right]$, with all-outgoing charges $(-1,+1,+1,-1)$ and $\mu^2 = 10^6$. The retained FKS convention cancels the auxiliary QES scale algebraically; these checks introduce no scale-varied amplitude evaluations.

Table 3: Universal QED double pole.

Point	Eligible fixed-QP worker	Predicted pole	Observed pole	Relative residual
1†	CutTools QP control	-7.02262×10^{-5}	-7.02262×10^{-5}	3.674×10^{-16}
1†	Private candidate TIRust QP	-7.02262×10^{-5}	-7.02262×10^{-5}	1.857×10^{-17}
20	CutTools QP control	-1.62777×10^{-4}	-8.13900×10^{-5}	5.000×10^{-1}
20	Private candidate TIRust QP	-1.62777×10^{-4}	-1.62777×10^{-4}	2.679×10^{-16}

Table 4: Universal QED single IR pole of the intended renormalized amplitude.

Point	Eligible fixed-QP worker	Predicted pole	Observed pole	Relative residual
1†	CutTools QP control	-1.44006×10^{-4}	-1.44006×10^{-4}	5.481×10^{-17}
1†	Private candidate TIRust QP	-1.44006×10^{-4}	-1.44006×10^{-4}	2.430×10^{-16}
20	CutTools QP control	-4.96751×10^{-3}	-2.33547×10^{-3}	5.299×10^{-1}
20	Private candidate TIRust QP	-4.96751×10^{-3}	-4.96751×10^{-3}	5.231×10^{-16}

† Point 1 rows are the selected point-20 workers' bootstrap, not additional independent physical samples. Two exactly duplicated answer orders are collapsed once. Current candidate CutTools controls match the displayed original CutTools rows exactly.

At point 20, private candidate TIRust QP agrees with the reconstructed double and single IR poles to about 2.68×10^{-16} and 5.23×10^{-16} respectively. Original CutTools QP differs by about 50.0% and 53.0%. This is conditional coefficient consistency; the finite coefficient and full-amplitude accuracy remain unresolved.

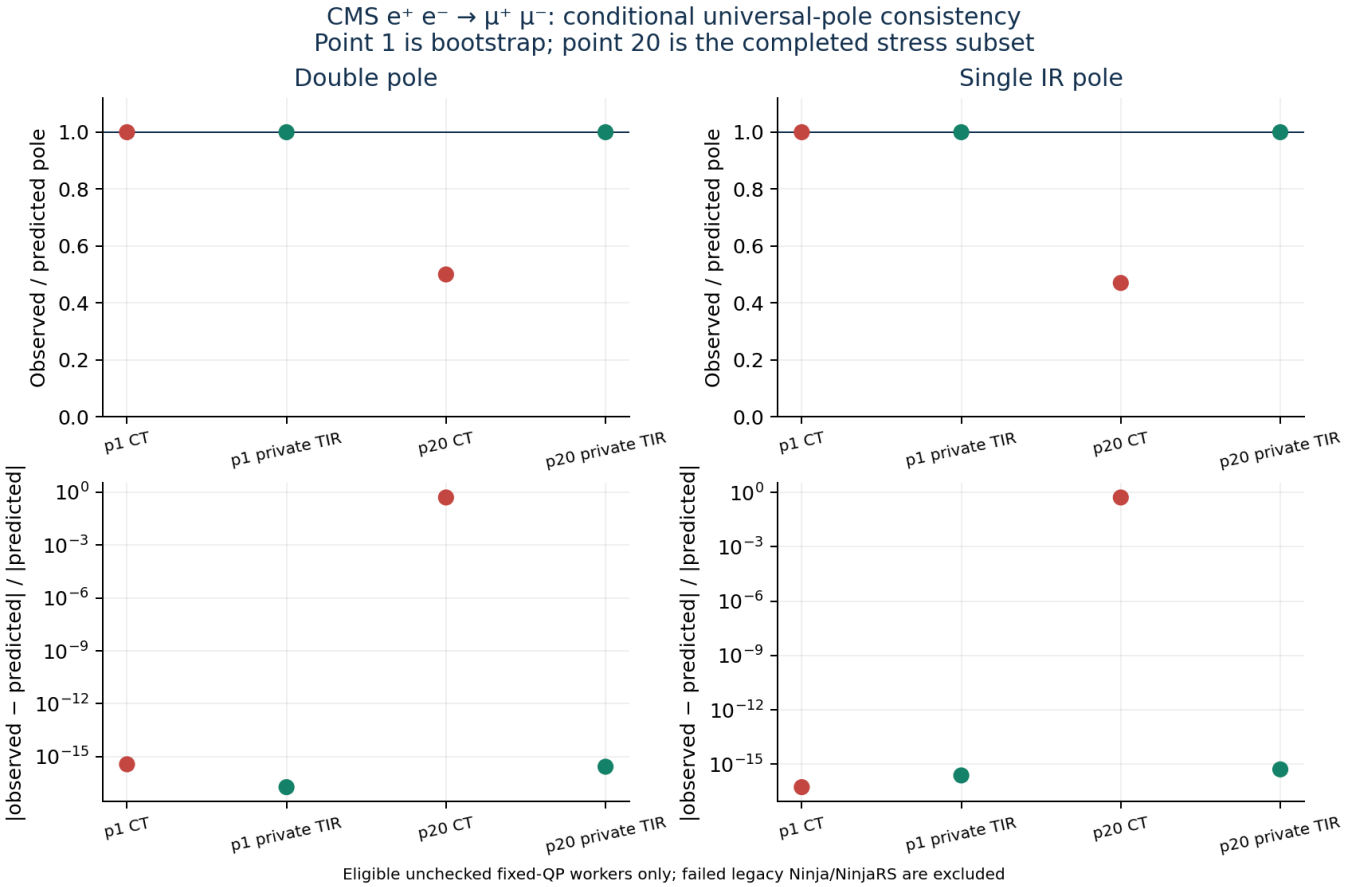


Figure 1: Observed/predicted poles and relative residuals for the eight eligible observations. Failed legacy probes are excluded, not assigned zero.

Assumptions that limit the conclusion

The coupling is reconstructed as $\alpha = \sqrt{2}G_F \frac{\left|\mu_W^2\left(1-\frac{\mu_W^2}{\mu_Z^2}\right)\right|}{\pi}$, $\mu_V^2 = M_V^2 - iM_V\Gamma_V$, with the existing `GFRedefinitionChoice=1` and the retained model's decimal pi literal. Parsing card values through binary64 is an explicit assumption: retained packets do not contain the card-reader/MP-widening implementation or observed runtime coupling bits. The original alpha and pole calculation uses 70-digit Decimal working arithmetic; independent pole-factor and scale-cancellation reviews use 90 digits. Neither measures native coupling precision or emulates every Fortran rounding step.

The single-pole calculation uses earlier closed group-3 QP trace leaves for points 1/20 and exact rational dot products of those words. Their original point-20 stress run remains failed. Matching cards, point streams and an unchanged momentum-source path justify the documented reuse; the current prototype made no new external-word capture. Restored QP leaf components must not be equated with the un-restored binary64 input stream. The prediction applies to the intended UV-renormalized full `ANS`, with counterterms and UV+IR selection. Inclusion of counterterms is source-linked; UV cancellation and every model counterterm were not separately tested. This is not a bare tensor `IR1` test.

Reproduction and remaining acceptance

The pure generator authenticates the four closed public indexes and every indexed payload, joins the selected eligible pole words to the closed parent, and preserves failed reference buffers without decoding or arithmetic. It does not re-execute the packets' broader dependency verifiers, rehash today's external binaries, or reparse all historical generated sources.

Public packet seals: double pole `1565ac55fe6a…`; single pole `99603d41d662…`; private amplitude `c04a73c86f6a…`; legacy references `248bd61e987a…`. Complete seals and selected leaf hashes are in `data.json`.

Local precise timing TODO — Hua-Sheng. Use identical local builds, points, models and cards; run at least seven alternating paired batches of about one second. Require the upper 95% runtime ratio bound to be ≤1.05 for every main process, with existing-backend regressions and native-DP/rescued cohorts reported separately. Current stage elapsed times are accountability records, not measurements satisfying that requirement.

Later matched-input QP comparison

Later evidence; the preceding three pages retain their earlier chronology. Two eligible private fixed-QP workers now agree at the same CMS point 20 in Born, finite, single- and double-pole outputs. The maximum retained `abs(a-b)/max(abs(a),abs(b))` is 2.014×10^{-16}. This is matched-input agreement, not an analytic finite oracle or an algorithmic winner. The private input-policy change is not promoted.

The closed comparison uses backend 8 (NinjaRS) and backend 9 (TIRust), one point-1 bootstrap and one point-20 measurement each: four outer calls and 104 native adapter calls. All native statuses are zero, both whole-worker eligibility gates and bootstrap controls pass, and post-call software DP/QP failure flags are clear. Fixed return 120 remains an unchecked evaluation.

What was matched and what changed

For all 52 paired native requests, the ordered shifts, full scalar-invariant matrix, complex squared masses, scale and canonical numerator words match exactly. Explicit routing/producer metadata and entry/use environment tokens also match. The ordered group-3 entries $(2, 4)$ and $(4, 2)$ actually use the certified direct spacelike invariant. QP interface storage is 128 bits with 113 precision digits; this does not assert every reducer's internal precision.

Backend 8 here uses a separate private input-policy derivative. It shares the candidate's unchanged certified invariant helper, retains a provably negative raw invariant where a qualified composite pair would otherwise be overwritten by the legacy zero branch, aligns the authoritative-upper symmetry policy, and evaluates the scale as `REAL(MU_R,KIND=KI_QNRS)**2`. Original legacy 8 retains its original helper and `REAL(MU_R**2,KIND=KI_QNRS)` route. No library, scalar cutoff, physical card or production backend is changed by this test. The earlier failed original-8/Ninja-6 reference probe remains failed.

Table 1: Point-20 retained binary64 outer-output words from fixed-QP evaluations. Both answer orders agree; exact duplicate orders are counted once.

Coefficient	Private NinjaRS 8	Private TIRust 9	ULP gap	Relative residual
Born	3.38643×10^{-2}	3.38643×10^{-2}	0	0
Finite	-3.44596×10^{-2}	-3.44596×10^{-2}	1	2.014×10^{-16}
Single pole	-4.96751×10^{-3}	-4.96751×10^{-3}	1	1.746×10^{-16}
Double pole	-1.62777×10^{-4}	-1.62777×10^{-4}	0	0

Born and double-pole words are bit-identical. Finite and single-pole outputs differ by one ULP of the retained binary64 word. All coefficients and both orders satisfy the existing regression criterion and the 10^{-8} target. Native success alone was not used to admit coefficient arithmetic.

Connection to the earlier pole audits and remaining scope

The current backend-9 output words match the prior eligible private candidate exactly: sixteen Born/Laurent words across bootstrap/measurement and two orders, plus four ancillary diagnostic words. Source/runtime-provider bindings are retained. Historical native input words were not captured, so this output parity does not retroactively measure equality of the earlier native inputs. The preceding conditional pole audits concern those same observed outputs; their coupling-reconstruction, restored-leaf and UV-cancellation limitations remain in force. No failed reference coefficient is selected or decoded.

Still unqualified. Two-reducer agreement does not establish independent full-amplitude truth, a general source/exporter fix, normal checked MadLoop stability/rescue, generic process/topology coverage or performance. The earlier incomplete stress stage is not reclassified. Precise local timing remains Hua-Sheng's TODO with the agreed per-process upper-95% ratio bound ≤1.05.

Closed run `52faa470d8be…`; comparison `af4b58a379ac…`; root review `9ca6601ab14a…`; prior/current word parity `5f9f4cc6c8d2…`. Full pins and the public evidence join are retained in the companion data/checker. Closed public evidence.